THE COKERNEL OF THE LAST DIFFERENTIAL OF A FLAT COMPLEX WITH BOUNDED TOR AMPLITUDE IS FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: each of the three steps, the removal of the derived tensor product, the middle exactness and the identification of the Tor group, is stated in the counted span.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space, and where applicable in chapter more-algebra for a ring. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here; the special-case twins are to be left uncounted by the lanes holding those chapters.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Tor amplitude of a complex of O-modules. Same chapter, the definition of tor amplitude in [a, b] as the vanishing of the cohomology of E^bullet tensor^L F outside [a, b]. Supplies the hypothesis that the complex has tor amplitude in [a, b], which is applied in the middle degree a - 1.
- Flatness of a module over a ringed site and exactness of tensorisation. Same chapter, the definition of a flat O-module. Supplies the first step, where flatness of the terms makes the naive tensor product compute the derived one.
- Long exact sequence of Tor and the computation of Tor by a flat resolution. Same chapter, Tor computed from a flat resolution of the first variable. Supplies the identification of Tor_1 of the cokernel with the middle homology of the displayed complex.

Source reference. Original lemma 08FG: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3360–3360. Same-chapter navigation pointer invoked at proof line 12428, where the author converts the vanishing of Tor_1 into flatness. The pointer only resolves that reference for the reader; the cited criterion is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{E}^\bullet$ be a bounded above complex of flat $\mathcal{O}$-modules with tor-amplitude in $[a, b]$. Then* $\mathrm{Coker}(d^{a-1}_{\mathcal{E}^\bullet})$ *is a flat $\mathcal{O}$-module.*

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Proof. As $\mathcal{E}^\bullet$ is a bounded above complex of flat modules we see that $\mathcal{E}^\bullet \otimes_{\mathcal{O}} \mathcal{F} = \mathcal{E}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{F}$ for any $\mathcal{O}$-module $\mathcal{F}$. Hence for every $\mathcal{O}$-module $\mathcal{F}$ the sequence

$$\mathcal{E}^{a-2} \otimes_{\mathcal{O}} \mathcal{F} \to \mathcal{E}^{a-1} \otimes_{\mathcal{O}} \mathcal{F} \to \mathcal{E}^a \otimes_{\mathcal{O}} \mathcal{F}$$

is exact in the middle. Since $\mathcal{E}^{a-2} \to \mathcal{E}^{a-1} \to \mathcal{E}^a \to \mathrm{Coker}(d^{a-1}) \to 0$ is a flat resolution this implies that $\mathrm{Tor}_1^{\mathcal{O}}(\mathrm{Coker}(d^{a-1}), \mathcal{F}) = 0$ for all $\mathcal{O}$-modules $\mathcal{F}$. This means that $\mathrm{Coker}(d^{a-1})$ is flat, see Lemma 08FG. $\square$